

Correct Now, Insufficient Later: Auditing Update Sufficiency in Context Compression

Guangzhe Zhang

Independent AI Researcher
twhite.zh@gmail.com

17 September 2026

Abstract

A memory can answer a current query correctly while discarding distinctions required by a later update. We investigate this failure with a paired-history audit: two histories have the same current answer, receive a shared future update, and require different subsequent answers. A pilot evaluates 24 history pairs across six synthetic mechanisms, 12 memory conditions, two repeats, and two model backends. A deterministic frontier selector obtains strict reveal accuracy of 96/96 on DeepSeek and 82/96 on GLM; a structured writer obtains 62 successes with one unresolved outcome and 56/96. The configured four-outcome joint contrast has finite-sample identification intervals of $[0.521, 0.542]$ and $[0.292, 0.313]$, not confidence intervals. A record-level audit distinguishes retained-state adequacy, response delivery, and answer-schema compliance without changing those original scores. It finds 26 and 25 well-formed but semantically wrong structured reveal memories, while all 14 GLM frontier reveal failures contain correct values in the wrong wrapper. Tombstone removal produces 16/16 exact replay failures in the targeted mechanism. Identifier renaming then exposes a separate flaw: original frontier late-reference adequacy falls from 8/8 to 94/320 transformed instances. We provide and test a label-equivariant repair, but it preserves only 2/8 original late-reference answers: eliminating a naming shortcut does not solve unknown future relevance. These results support a scoped evaluation methodology and reproducible failure analysis, not general superiority of the repaired algorithm. Paid pilot evidence, retrospective diagnostics, and new offline tests are reported separately; no independent held-out or natural-task validation is claimed.

Keywords: context compression; update sufficiency; agent memory; temporal validity; metamorphic testing; reproducibility.

1 Introduction

Context compression selects information for future use. A short memory may accurately describe the present and still omit an older version, a persistent revocation, or a dependency that becomes decisive later. Consider two histories whose current value is X . Their preceding valid values are A and B . After the same event revokes X , the correct answers diverge. A memory that retained only X cannot recover both alternatives without another source of information. Current-answer agreement is therefore an inadequate stand-alone test of a memory’s usefulness under updates.

This observation is related to longstanding predictive-state reasoning, which represents history through its implications for future observations, rather than a new discovery that temporal information matters

(Littman, Sutton, and Singh 2001). Contemporary memory evaluations also include temporal reasoning and knowledge updates (Wu et al. 2024, 2026). The narrower question here is methodological: what can be learned by holding the current answer fixed, applying the same future to different histories, and inspecting both the stored evidence and the eventual response?

We study this question in CCA v4, a small symbolic event-log environment. Its paired histories agree on the current query but differ in latent evidence. A revealing future makes that evidence necessary; an overriding future instead installs a common new answer. The design tests whether a memory can both preserve earlier distinctions and accept legitimate replacement. We compare a transparent frontier selector, targeted ablations, model-written memories, simple selection rules, and separately labeled archive-access references.

An initially favorable performance comparison does not by itself identify a compression mechanism. A model-written memory can fail to arrive, exceed the size cap, contain the wrong state, or be correctly interpreted but rendered in a rejected schema. A deterministic selector avoids some of these paths by construction. Comparing the final scores therefore measures the implemented systems, not a pure content-selection effect. In addition, a symbolic generator can reward arbitrary naming conventions. Those conventions must be perturbed before original-sample success is interpreted as robust selection.

This study consequently makes three scoped contributions. First, it formalizes a matched-current, shared-future audit and an all-four-future-outcomes endpoint. Second, it reconstructs the pilot’s complete result grid and gives a disjoint record-level diagnostic analysis without deleting unsuccessful writes or redefining the primary metric. Third, it introduces a concrete metamorphic contract for the reference selector, implements a minimal label-equivariant repair, and reports its negative as well as positive offline results. The repair removes lexical identifier dependence; it does not overcome bounded memory’s uncertainty about later references.

The evidence is organized into three layers throughout: **P**, the original paid pilot; **D**, retrospective analysis of those fixed artifacts; and **R**, new offline repair tests on transformations of the same source histories. Neither D nor R is an independent confirmation of P. Both backends use the same 24 source pairs, and the dataset is explicitly labeled development data. There is no claim of natural-domain, long-horizon agent, or state-of-the-art algorithmic performance.

2 Related work and contribution boundary

2.1 Compression and memory evaluation

ACON optimizes natural-language compression guidance using trajectory failures, and The Complexity Trap demonstrates that simple observation masking can compete with LLM summaries in its evaluated software-agent setting (Kang et al. 2025; Lindenbauer et al. 2025). These results motivate stronger baselines than an arbitrary short summary. Our local structured writer, tail selector, and lexical ranker do not reproduce those systems’ best configurations, so outperforming them cannot establish superiority over either published method.

TRACE evaluates compression boundaries through closed-loop continuations from a shared environment state (Min et al. 2026). Our pairing is different: complete historical states intentionally differ, but their current query projections agree. A shared future then exposes a distinction that a compressor might have lost. This is complementary to same-state execution evaluation; a synthetic value resolver does not replace a tool-using agent. The Compaction Cliff further motivates protecting different kinds of information under repeated compression (Zerhoubi, Mitrovic, and Granitzer 2026). Our two-chunk futures do not

reproduce its long-running retention setting.

MEMAUDIT is the closest neighbor for attribution. It evaluates budgeted memory writing with explicit evidence requirements, validity state, representation choices, and an exact package oracle (Bhargava and Barrento 2026). We do not claim novelty for retaining tombstones, separating writing from reading, or using exact semantic checks. Our interpreter evaluates a *given* event memory; it does not optimize the best package under a budget. The additional design studied here is the matched-current pair, its two shared futures, and a concrete metamorphic audit of the resulting reference selector.

2.2 Temporal memory, reasoning, and validation

Zep, A-MEM, and Mem0 maintain and evolve memories using different architectures (Rasmussen et al. 2025; Xu et al. 2025; Chhikara et al. 2025). LongMemEval and LongMemEval-V2 already include update-sensitive capabilities; AppWorld evaluates executable interactive agents (Wu et al. 2024, 2026; Trivedi et al. 2024). None is numerically ranked by this pilot. We therefore avoid the broader claim that existing memory evaluation is uniformly static or that symbolic retention implies real-world task success.

Reasoning-enabled summarizers are not uniformly superior in prior dialogue-summarization experiments (Jin et al. 2025). The present H/off writer comparison is weaker evidence than a general reasoning study: it uses backend-specific controls, a common output ceiling rather than matched compute, and a joint endpoint near the floor. Its zero joint contrast is not an equivalence result.

Our relabeling checks instantiate metamorphic testing: known relations between transformed inputs and outputs test a system even when a new independent oracle is expensive (Chen et al. 2018). We claim neither to introduce metamorphic testing nor to make transformation counts equivalent to new tasks. Finally, excluding empty or malformed memories after treatment would change the comparison population. The general danger of post-treatment selection is well established (Montgomery, Nyhan, and Torres 2018). We retain all planned cases and use diagnostic cross-tabs, not a mediation estimate based on selected survivors.

3 Update sufficiency and evidence levels

3.1 Matched-current histories

Let h be a history, q a query, u a future event sequence, and $A(h, q)$ the exact answer under public semantics. Concatenation is $h \oplus u$. A pair (h_0, h_1) is currently equivalent for q when

$$A(h_0, q) = A(h_1, q).$$

It is future-distinguishable relative to an update family $\mathcal{U}$ when

$$\exists u \in \mathcal{U} : A(h_0 \oplus u, q) \neq A(h_1 \oplus u, q).$$

This is equality of current task outputs, not equality of hidden historical state. A compressor writes $m_0 = C(h, q; B)$ within budget B . On the strict-memory track, subsequent state is $m_{t+1} = U(m_t, \delta_{t+1}, q; B)$: the updater receives only the charged preceding memory and the new chunk. A reader returns a value from the resulting memory. Sufficiency is relative to declared queries and updates; it is not a promise to answer arbitrary future questions from bounded storage.

Observation 1 (collision obstruction). If two currently equivalent histories have identical compressed states, downstream side information is identical, and a shared future requires different target answers, no deterministic updater-reader composition is correct on both. If the compressed-state distributions are identical and the history variant is sampled uniformly, even a randomized downstream composition has mean correctness at most $1/2$.

Proof. The downstream input, or its distribution, is the same. A deterministic output cannot equal two distinct targets. For a randomized output, the probabilities of those mutually exclusive targets sum to at most one. Averaging over the balanced variants gives the bound. An archive, variant label, or other distinguishing side channel would violate the premise. This elementary argument is a diagnostic motivation, not a new general information-theoretic theorem.

The converse is false. Distinct memory strings can differ only in useless wording; an adequate memory can still be misread. In the pilot, the latest-only early states collide exactly in 12 of the 24 pairs, but further failures occur without literal collision. Both output evaluation and evidence inspection are needed.

3.2 What is and is not identified

The implemented arm a affects selected information, generated text, and delivery behavior. A schematic dependency is

$$a \longrightarrow (\text{content}, \text{delivery}) \longrightarrow m \longrightarrow r \longrightarrow S_q(r),$$

where S_q is the strict answer contract. Reader sampling and backend behavior also affect r . A score difference between deterministic frontier and a model writer is consequently an end-to-end contrast between their implemented pipelines. Holding the backend name fixed does not equalize writer reliability, output length, or reasoning consumption.

For event-schema memories we additionally compute $E_q(m)$, the exact value obtained by a separately implemented interpreter. If parsing fails, E_q is unavailable, not automatically incorrect: a nonconforming string can still contain an answer usable by an LLM. For completed responses we define a diagnostic $S_q^{\text{bare}}(r)$ that also accepts an exact bare query-value map. The original S_q is never replaced. This yields three distinct measurements: usable strict output, adequacy under the explicit event interpretation, and sensitivity to one formatting convention.

These cross-classifications localize observable failure patterns but do not identify the causal proportion of an arm gap due to content. They neither condition the primary denominator on successful writing nor infer what a missing response would have contained. Claims from P, D, and R retain their distinct statuses even when all three agree.

4 Pilot design and implementation

4.1 Histories, semantics, and futures

The development dataset has 24 pairs: four each for rollback, expiry, choice switch, late reference, tombstone replay, and a mixed alias/expiry mechanism. Both models evaluate the same 48 variants. Each prefix includes 32 distractor records. Initial event order is shuffled. The early writer receives the target keys and public event semantics, but the reference task constructor excludes the future text, gold answer, variant label, and mechanism label. Thus the setting is target-aware and future-hidden, not query-agnostic. Full wire-level enforcement cannot be attested from the exported artifacts alone.

Events form an append-only log with unique identifiers. Identical replay is idempotent; conflicting content with the same identifier is invalid. A **revoke** permanently invalidates its target identifier, including future replay. The clock is the maximum retained **tick.now**, initially zero; expiry is inclusive. For each key, resolution selects the greatest-revision non-revoked, unexpired **set**, **copy**, or **choose** event. A **set** yields a string; a **copy** resolves another key; a **choose** resolves a selector and one of two referenced keys. Missing keys and cycles return null. A dangling dependency of a valid selected record returns null rather than falling back to an older version. Unique key-revision pairs eliminate revision ties.

Each prefix is first read on the current query, then forks from the same early memory into two futures. The *reveal* future exposes the hidden historical distinction: it revokes the latest value, advances time, switches a selector, introduces a previously irrelevant reference, or replays an invalidated record. The *override* future installs a new highest-revision answer common to both variants. Each future has two chunks: the meaningful operation first and a distractor second. This is not a long sequence of independent reversals.

Two repeats are planned for every arm and source history. For deterministic memories, only model responses can vary; generated writers can also vary their memories. Repeated calls, future branches, and history variants are nested within the source pair. The configured runner randomizes scheduling, but exact deployed order requires the absent run ledger and lock files.

4.2 Conditions, storage access, and fairness

All returned non-full memories are capped at **1,200 serialized UTF-8 bytes**, including identifiers, controls, JSON syntax, and ordering. Whole-record packers skip records that do not fit. The cap is not 1,200 tokens. **full** is uncapped. Strict-memory methods see preceding memory plus new events only; archive methods may inspect accumulated raw history again after updates. An archive is an explicit additional capability, not free bounded persistence.

The eight strict-memory conditions are **tail**, **type_pin**, **frontier**, **frontier_latest**, **frontier_no_tombstone**, **prose_high**, **prose_off**, and **structured_high**. Tail packs complete records in reverse arrival order. Type-pin prioritizes clock and revocation events. The prose writers recursively rewrite free text; structured H requests an events object and explicitly asks for old valid versions, dependencies, expiry, and tombstones. The four archive references are full history, lexical BM25 selection, dependency-closure selection, and fresh direct H-policy prose writing from accumulated history. Appendix A reports every arm.

Original frontier normalizes the previous records and incoming events, filters invalid or expired value records, and collects revocations plus the latest maximum clock. It computes the transitive key closure through *all live versions* of the query targets. Controls are ordered first, then live events by target membership, closure membership, revision, and identifier. Whole-record inclusion is tested against the exact JSON byte cap. Residual capacity can preserve non-closure events. This is a heuristic, not an optimal-memory oracle or a guarantee that every control record fits.

Latest-only restricts live records to the largest revision per key before closure and packing. It is not literally one scalar value: referenced records may remain. No-tombstone first uses revocations to filter current live records but then omits those controls from persisted memory. Replayed invalid identifiers can therefore become active after the deletion marker has been forgotten. These ablations manipulate explicit persistence rules; they are not separately optimized architectures.

4.3 Backends, delivery, and output scoring

The exported profiles request `deepseek-flash` and `glm-5.2` through Chat Completions. Returned names match their respective allowlists for every nonmissing reader receipt. This checks exported aliases, not immutable weights or router behavior. DeepSeek H enables thinking and requests high effort; GLM H enables thinking without an explicit effort setting. Off disables thinking for that writer, not for its reader. Equal policy labels therefore do not establish cross-vendor compute equivalence.

Writer and reader output allowances are 4,096 tokens, with a 600-second configured timeout. A documented DeepSeek accounting-boundary amendment accepts one returned receipt metered at 4,097 completion tokens. It is still an incomplete, empty, incorrect reader response. It is not an extra trial or a rescued success.

The memory acceptance policy is consequential. Missing infrastructure responses propagate unknown outcomes. An over-cap returned memory is replaced by an empty string; naturally empty output remains empty. An incomplete response within the cap is retained and flagged. There is no universal rollback to the last valid memory and no automatic retry-until-success policy. Thus generated memories face delivery failure modes that deterministic selectors avoid.

Strict reader success requires a stop-finished response of exactly `{"values": {query_key: string_or_null}}`, with precisely the required keys and values. Invalid structure, incorrect values, and incomplete output score zero. Infrastructure missingness stays NA. This is the preserved operational metric, even where a less strict consumer could use the output.

4.4 Endpoint, missingness, and uncertainty

Let Y_{ipavb} be strict success for source pair i , repeat p , arm a , variant v , and branch b . The configured joint endpoint is

$$J_{ipa} = \prod_{v \in \{0,1\}} \prod_{b \in \{\text{reveal}, \text{override}\}} Y_{ipavb}.$$

Current performance is reported separately and does not filter eligibility. An arm has 48 pair-repeat blocks, but only 24 source-pair clusters. We average repeats within each pair before averaging pairs. This endpoint is intentionally demanding: success requires retaining history-sensitive information *and* accepting a replacing update for both variants.

For an incomplete block, any observed zero identifies $J = 0$. Otherwise, a block with an unresolved constituent has bounds $[0, 1]$. Averaging block lower and upper values yields $[L_a, U_a]$. The frontier-minus-baseline contrast is bounded by

$$\Delta \in [L_{\text{frontier}} - U_{\text{baseline}}, U_{\text{frontier}} - L_{\text{baseline}}].$$

These are finite-sample identification bounds over unobserved outcomes, not confidence intervals or tests of a population null. All planned denominators remain fixed, without a missing-at-random assumption.

The archived analysis also resamples pairs within each of the six fixed families, retaining their repeats and branches: 2,000 draws with seed 20260911. It reports the 2.5th percentile of lower endpoints and 97.5th percentile of upper endpoints, following the resampling principle of the bootstrap (Efron 1979).

With only four pairs per family, this is an approximate conditional endpoint envelope, not a calibrated simultaneous confidence set for unfamiliar mechanisms.

The supplied configuration names frontier versus structured and prose off versus H across three planned model slots; only two slots have results. The unused model is not silently removed to obtain a smaller multiplicity family. Holm correction does not repair invalid underlying tests (Holm 1979). No confirmatory frontier p-value is reported. Configuration intent is not evidence of externally timestamped preregistration.

5 Results: paid pilot and retrospective attribution

5.1 Complete exported grid, bounded provenance

Both backends contain all 3,456 planned outcome records, with no duplicate or extra identifiers. DeepSeek has one NA and GLM three. Independent rescoring reproduces every original score and failure class. Dataset and gold hashes match the reference kit, and a separately implemented interpreter verifies every supplied gold answer.

The exported evidence supports **5,181 done operations on DeepSeek and 5,180 on GLM**, or **10,361 done**, not 10,362. Including four missing operations gives 10,365 recorded operations below the 10,368 ceiling. Three planned downstream calls were not dispatched after upstream missing writers. The boundary-settlement receipt is already present in the outcomes and is not counted twice. Complete exported grids establish completion of the recorded analysis surface, not live-process exit or invoice reconciliation.

We additionally replay the eight deterministic reference conditions from source histories, reconstructing 1,152 distinct branch memories and comparing every corresponding repeat/backend export. **All 4,608 exported memories match byte-for-byte**. This strengthens functional reproducibility for those states. It does not attest the complete deployed source, model-written requests, or absence of external side channels.

5.2 Current success and update success diverge

Table 1. Selected original strict results (P), with fixed denominators. DS = DeepSeek. Current and reveal columns have 96 planned reads; joint columns have 48 pair-repeat blocks within 24 pairs. An asterisk indicates one unresolved case or block, not an added success. Full archive is not storage-equivalent to strict memory.

Condition	DS current	DS reveal	DS joint	GLM	GLM reveal	GLM joint
				current		
Frontier	96/96	96/96	45/48	78/96	82/96	28/48
Latest-only	96/96	32/96	15/48	84/96	23/96	5/48
No tombstones	95/96	79/96	38/48	82/96	65/96	20/48
Structured	76/96	62/96*	19/48*	64/96	56/96	13/48*
H						
Full archive	96/96	95/96	46/48	83/96	75/96	23/48

The clearest dissociation is DeepSeek frontier versus latest-only. Both score 96/96 on the current branch; reveal falls from 96/96 to 32/96 when older live records are removed. On GLM, latest-only has higher

current strict accuracy, 84/96 versus 78/96, but lower reveal accuracy, 23/96 versus 82/96. Current-answer performance alone can therefore prefer a much less update-capable memory on these histories. This is not a claim that static quality and updateability are statistically independent.

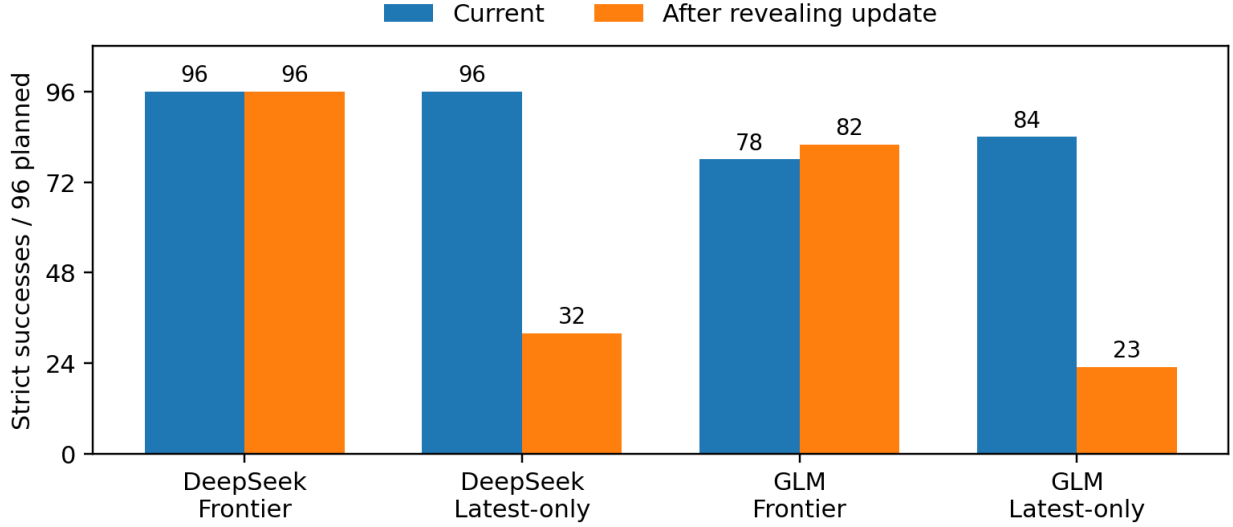


Figure 1: Current and reveal strict successes on the original paid pilot. Shared source pairs and repeated reads are not independent tasks. Numerical values are also in Table 1.

Table 2. Frontier minus structured H: original fixed-sample identification bounds and approximate conditional bootstrap envelopes. Joint and reveal are different endpoints; none of these bounds establishes external-domain performance.

Endpoint	Backend	Identification interval	Approx. bootstrap envelope
Joint	DeepSeek	[0.520833, 0.541667]	[0.395833, 0.646354]
Reveal	DeepSeek	[0.343750, 0.354167]	[0.260417, 0.447917]
Joint	GLM	[0.291667, 0.312500]	[0.104167, 0.479167]
Reveal	GLM	[0.270833, 0.270833]	[0.156250, 0.385417]

The prominent joint intervals, [0.520833, 0.541667] and [0.291667, 0.312500], must not be presented as reveal-only gaps. The reveal difference is 34.38–35.42 percentage points on DeepSeek and 27.08 points on GLM. Frontier’s strict joint score is 45/48 on DeepSeek, below full history’s 46/48; on GLM it is 28/48 versus 23/48. Neither pattern establishes a general advantage over complete raw-history access.

Pair-level heterogeneity is retained. Against structured H, DeepSeek has 16 definitely positive pair-mean joint contrasts, seven ties, and one negative; GLM has 12 positive, eight ties, three negative, and one unresolved contrast. A leave-one-family-out descriptive audit keeps the joint lower endpoint positive for every omission: its minimum is 0.45 on DeepSeek and 0.20 on GLM. These checks show the aggregate does not depend on only one family, but remain reuse of development data, not replication on omitted-family training or unseen-family evaluation.

5.3 A joint content-and-response diagnostic

Table 3 partitions all 96 planned reveal records for each selected arm/backend, without changing their strict scores. “Unparsed” means that the event-schema interpreter cannot evaluate the retained memory. “Wrong log” means a valid log resolves to the wrong value. The last three columns all have a semantically correct event log but differ in reader output. These categories are diagnostic observations, not estimated causal components.

Table 3. Retrospective reveal cross-classification (D). Columns are mutually exclusive and sum to 96. Wrapper-only failures satisfy the exact bare-map sensitivity rule. Wrong-log cells in these selected rows have no strict successes; the implementation nevertheless preserves that possible discordance as a separate category.

Model	Condition	NA	Unparsed	Wrong log	Correct + pass	Wrapper only	Other reader
DS	Frontier	0	0	0	96	0	0
DS	Latest- only	0	0	64	32	0	0
DS	No tomb- stones	0	0	16	79	1	0
DS	Structured H	1	7	26	62	0	0
GLM	Frontier	0	0	0	82	14	0
GLM	Latest- only	0	0	64	23	8	1
GLM	No tomb- stones	0	0	16	65	14	1
GLM	Structured H	0	16	25	49	6	0

This partition sharpens attribution in two directions. First, structured H has 26 valid but wrong reveal logs on DeepSeek and 25 on GLM. Its problems are not confined to missing or syntactically malformed outputs. Second, all 96 frontier reveal memories are adequate under the exact interpreter on both backends. The 14 strict GLM frontier failures are therefore not cases of missing event evidence: all contain the correct bare value map but omit the required wrapper.

The event parser is not a universal semantic judge. Of GLM’s 16 unparsed structured reveal memories, seven still yield strict-correct reader answers. Those strings must not be labeled “information destroyed” merely because they are not valid event envelopes. Similarly, a wrong retained state and an accidentally correct answer would remain a discordant observation, not be rewritten to fit a neat failure narrative. All 6,912 result rows are assigned exactly once by the audit.

The alternative reader rule S_q^{bare} accepts only an additional stop-finished JSON object with exactly the query keys and correct string/null values. It does not remove code fences, infer keys, complete truncated text, or invoke an LLM judge. Applied uniformly to all arms, it changes GLM frontier reveal from 82/96 to 96/96 and structured H from 56/96 to 63/96; DeepSeek stays 96/96 and 62 successes with one NA. The primary operational scores are untouched. Consequently, the smaller strict GLM gap cannot be interpreted directly as weaker content retention or value inference.

Under this sensitivity frontier joint success becomes 48/48 on DeepSeek and 45/48 on GLM. The

corresponding structured joint counts are 19 and 17 known successes, each with one unresolved block. These are post-hoc diagnostic results, not replacement primary effects. Their purpose is to identify answer-contract dependence for a future frozen evaluation.

5.4 Localized ablations and delivery failures

The event interpreter resolves all original frontier, full, and closure-archive memories correctly. Latest-only resolves 32/96 reveal cases; no-tombstone resolves 80/96. The latter’s 16 failures are exactly the tombstone-replay mechanism: forgotten revocations allow invalid intermediate records to reappear. Actual strict reveal success in that stratum is 0/16 under no-tombstone on both models, versus 16/16 and 15/16 for frontier. This supports the specified persistence mechanism without requiring a language-model judge.

It would nevertheless be inaccurate to say the entire no-tombstone arm collapses. Its overall reveal scores are 79/96 and 65/96. Likewise, latest-only remains effective on choice-switch and original late-reference instances. Each family has four independent source pairs; sixteen repeated/variant outcomes are not sixteen independent demonstrations. Appendix A gives the complete mechanism table.

Delivery remains a substantial competing explanation for comparisons with generated writers. DeepSeek prose H has 115 raw-empty writes among 480 completed operations; direct H has 157 among 288. GLM has 85/480 and 47/288, respectively. Structured writers have 28 and 19 raw-empty returns. Empty and incomplete columns overlap and are not summed (Appendix B). The acceptance policy can also empty an over-cap result. Thus the pilot has not structurally eliminated empty-summary failures.

The H/off joint contrast is zero because DeepSeek has one passing joint block in each prose condition and GLM has none. Their reveal counts are not equal: 15 versus 14 on DeepSeek and 11 versus three on GLM. A floor-limited joint endpoint, equal output allowances, and different visible-output delivery cannot establish that reasoning policy has no effect. Neither discarding empty runs nor giving only one arm unlimited retries would repair the original comparison.

6 Metamorphic audit and a bounded reference repair

6.1 Identifier relabeling exposes an incidental shortcut

Original frontier uses identifier lexicographic order to resolve priority ties. In a post-hoc symbolic stress test, we replace every event identifier and key name with a unique random ASCII string of the same length, consistently rewriting references and query keys. Values, revisions, time, dependency topology, event order, and the 1,200-byte budget are unchanged. Full-history answers are independently verified after every transformation. There are 40 fixed renamings per source pair; the transformed instances are not new independent samples.

Original frontier late-reference adequacy drops from 8/8 original variants to 94/320 renamed variants. Each of the other five mechanisms remains 320/320. A previously irrelevant record is retained only when residual capacity and the lexical tie-break favor it. Closure archive continues to resolve all late-reference answers because it can access raw history after the new reference appears; that result relies on a different storage capability.

This is evidence of sensitivity to irrelevant naming, not proof that all earlier-version retention is ineffective. It also makes unchanged-generator expansion insufficient: more pairs with the same advantageous naming convention would not test the exposed failure.

6.2 A label-equivariant selector, not an improved paid result

We implement **frontier-alpha** after the pilot. It keeps the original control selection, all-version dependency closure, whole-record byte accounting, and strict update interface. The sole priority change removes lexical identifiers from both control and value tie-breaks. Equal-priority items preserve the order in which they occur in the current normalized input. That order is carried by the budgeted event array itself; there is no uncharged arrival-rank table or raw archive.

Let π consistently rename the event identifiers and key namespace without changing values, revisions, timestamps, equality relations, record order, or the serialized size of any record. For the canonical JSON encoding, the checked family uses same-length ASCII labels. The relevant contract is

$$C_\alpha(\pi(h), \pi(q); B) = \pi(C_\alpha(h, q; B)),$$

where the right-hand side includes decoding and canonically serializing the renamed event memory. The analogous property holds for each update when its previous memory and incoming chunk are renamed consistently.

Proposition 2 (label equivariance). Under those premises, frontier-alpha satisfies the contract at initial compression and at every strict-memory update, including when records are skipped because of the byte cap.

Proof. Normalization, identifier equality, revocation membership, live-version selection, target membership, and dependency closure all commute with a bijective renaming. Alpha’s priorities contain no lexical label comparison. Equal priorities preserve corresponding input positions, so the ordered candidate lists correspond. Every candidate has the same encoded length before and after renaming; induction over the packing loop therefore gives identical inclusion decisions. The resulting event arrays correspond under π . Decoding the preceding array and appending the correspondingly renamed next chunk preserves the premises, completing induction over updates. A cap below the 13-byte empty event envelope is rejected on both sides, rather than silently returning an invalid memory.

The premise concerns **serialized length**, not merely the number of Unicode characters. Escaping or different UTF-8 lengths can change capacity and invalidate a naive relabeling test. The proposition says nothing about permutation invariance, unknown future keys, optimal packing, or LLM reader performance.

6.3 The repair removes label dependence but not future uncertainty

We run both selectors on identity inputs, 40 same-length renamings, and 40 prefix permutations per pair. The permutations are shared by the two variants, preserve full-history answers under the event semantics, and do not alter the future. All intermediate memories are checked against the byte cap. The audit executes 13,440 stage-level alpha equivariance comparisons, with zero violations. These checks reuse source histories and repeated early states; their count is not a statistical sample size.

Table 4. Late-reference exact retained-state adequacy after reveal (R), at 1,200 bytes. This is an offline interpreter result, not a model score. Each row derives from the same four source pairs; 320 is eight variants times 40 transformations.

Offline input condition	Original frontier	Frontier-alpha
Original instances	8/8	2/8
Same-length renaming	94/320	80/320
Prefix permutations	320/320	78/320

On the original late-reference instances, alpha retains only 2/8 needed values. Under 40 renamings it retains exactly the corresponding 80/320, as equivariance requires. Under prefix permutations it retains 78/320. Original frontier remains 320/320 under these permutations because its lexical priority is unchanged. Each other family is 8/8 at identity and 320/320 under each transformation for both selectors. Thus protecting versions and tombstones survives these particular perturbations, whereas unknown late relevance remains unresolved.

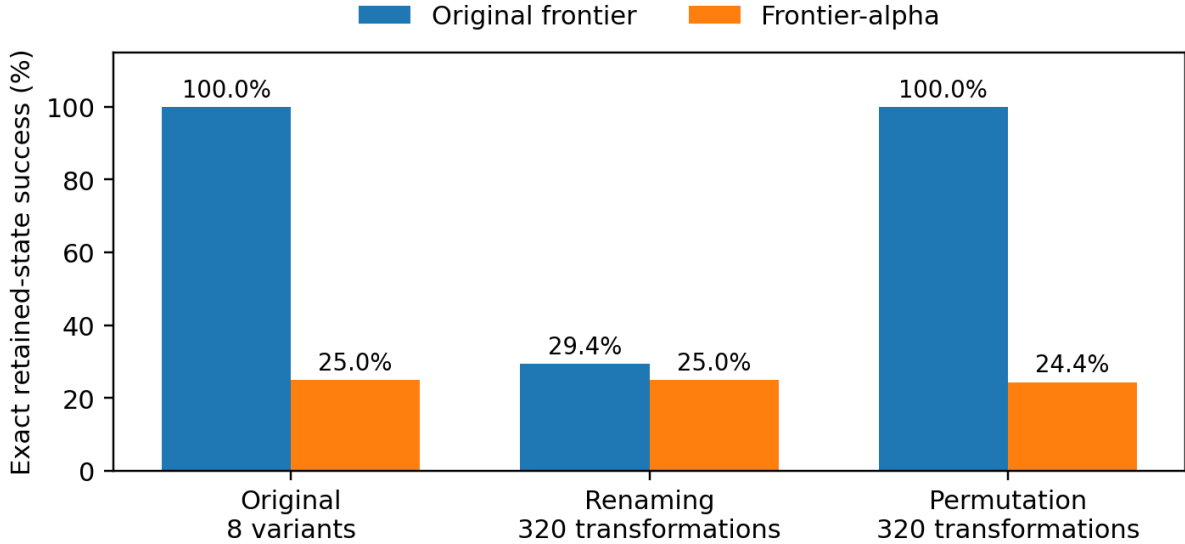


Figure 2: Late-reference retained-state adequacy. The repaired selector removes naming dependence without improving adequacy. Transformation counts reuse four source pairs and do not supply independent trials or error bars.

This negative repair result is scientifically useful. It separates an avoidable nuisance dependence from a substantive capacity/relevance problem. An invariant algorithm can be consistently inadequate; a high-scoring algorithm can exploit an incidental generator convention. Neither score alone validates a general compression mechanism.

A six-cap offline sweep further illustrates the boundary. On the original eight late-reference variants, alpha resolves 2/8 at 600, 900, 1,200, and 1,800 bytes, 4/8 at 2,400, and 8/8 at 4,800. These values are descriptive, not a tuned cap recommendation. Larger capacity can preserve more unrelated evidence, but does not make the fixed-budget selector universally sufficient.

Observation 3 (arbitrary late-query limit). Suppose a history contains N independently chosen values, each from an alphabet of size d , and a future can ask for any indexed value. With no side channel, a deterministic finite-state memory that answers every such query exactly must distinguish all

d^N assignments. If its representation has at most 2^b states, then $b \geq N \log_2 d$.

Proof. If fewer states encode all assignments, two different assignments collide. They differ at an index; querying that index requires different outputs from identical memory and side information. This contradicts exact recovery. This elementary counting argument is not a tight bound for our JSON packer, nor an impossibility claim about likely queries or archive-assisted systems. It makes the declared future family and storage budget explicit.

7 Reproducibility and validation scope

The artifact distributes sanitized outcome exports, source-data hashes, matching histories and gold values, the reference selector code, the alpha repair, all analysis programs, generated tables, and tests. No API call is needed to reproduce D or R. The paid data are kept unchanged. New scripts produce separate diagnostic and repair files rather than overwriting the original outcome labels.

The strongest executable checks are functional rather than historical: exact gold replay, all original scores, the complete grid, distinct operation accounting, 4,608 byte-identical deterministic memories, and alpha’s checked transformation contract. None reconstructs missing writer wire payloads or a server-side model snapshot. The result archive lacks the full live SQLite ledgers, run-lock manifests, some raw writer texts, and complete deployed-source attestation. Returned aliases are not immutable weights; a model-name amendment must not be interpreted as proof of unchanged underlying computation.

Known deduplicated receipts report 4,597,572 input and 4,196,900 completion tokens for DeepSeek, and 4,412,518 input and 5,623,634 completion tokens for GLM. Unknown usage for missing operations is not included. The artifacts do not establish complete currency cost, cache billing, or equal compute across vendors. We make no latency or monetary-efficiency superiority claim.

The evidence supports three bounded conclusions. On these development histories, current-answer success can conceal future-update failure. Under the specified semantics, forgetting versions and revocations causes localized content failures. Finally, response formatting and incidental label order materially affect apparent performance. The evidence does **not** establish a robust general compressor, external-benchmark superiority, an equivalent H/off writing policy, a repaired model result, or a pure semantic effect of frontier versus equally reliable structured writing.

The repair addresses a reproducible implementation dependency; the cross-tabs correct attribution language; fixed denominators and evidence levels correct validation claims. They do not manufacture missing external evidence. A complete manuscript can report an honest exploratory mechanism study while still requiring additional work for stronger claims.

8 A prospective confirmation contract

A confirmatory study should not simply increase the unchanged generator to 192 pairs. Its implementation and measurement contracts must first be repaired on development material, and all resulting changes require new experiment identities. The following design is proposed, **not executed** in this paper.

First, separate representation selection from delivery. Compare deterministic alpha, a structured writer, a simple whole-record baseline, and targeted ablations under explicit storage parity. A common deterministic delivery layer should validate the envelope, enforce the byte cap, and record whether the candidate or a bounded fallback was committed. The original reject-to-empty policy and the validated-commit policy should be crossed with applicable methods, rather than improving only the favored arm. A fallback must

use only charged prior memory and the new chunk, never an uncharged archive or future labels. Extra requests and fallback frequency are outcomes and costs, not grounds to discard a trial.

Second, use a fixed semantic interface without erasing operational failures. Before running confirmation, declare strict response validity as one endpoint and exact value accuracy as a separate secondary endpoint; specify the accepted schemas without reference to new outcomes. Persist raw output and finish status. A length-finished or transport-missing response is not silently repaired into a completed success. Paired source histories, complete outcome grids, missingness bounds, and source-level clustering remain fixed.

Third, distinguish **within-generator confirmation** from **external validation**. New seeds alone supply new instances of an old mechanism. Independently authored mechanisms and source-separated natural or executable tasks are needed for transfer claims. Freeze task sources, retrieval permissions, query visibility, budgets, randomization, and stopping criteria before exposing outcomes. Renaming and order tests should be included prospectively, not selected afterward because a candidate passes. No result from the revised development audit can be relabeled as held-out evidence.

Finally, freeze the primary contrast, model identities or documented alias limits, and the multiplicity family before choosing a confirmatory sample size. Estimate cost and precision on development data, not on sealed-test success. LongMemEval-V2 and AppWorld offer complementary evaluation settings, but adapting them is a separate study, not a result of this pilot (Wu et al. 2026; Trivedi et al. 2024). The supplied confirmation contract has execution disabled and does not initiate paid work.

9 Limitations

The dataset has only four pairs per mechanism, and mechanisms share one synthetic grammar. Its selector understands that grammar directly. A successful interpreter reconstruction therefore does not establish useful compression of natural conversations, repositories, or partially observed environments. Two update chunks are not a long-horizon stress test. Repeated calls provide operational observations but do not create independent task diversity.

Our structured baseline is informed about relevant event types, but is neither delivery-matched to deterministic selection nor a faithful reproduction of the strongest published systems. The diagnostic tables cannot recover its counterfactual performance under repaired writing. The event interpreter has deliberately narrow coverage; seven GLM structured successes from unparsed memory demonstrate why parser failure must not be equated with semantic loss. Format sensitivity and selector repair were developed after seeing the pilot.

Alpha establishes label equivariance only under its stated transformation premises. It remains sensitive to record order and unknown late relevance. Its tests reuse the original pairs, use an exact symbolic reader, and have no live backend evaluation. A zero violation count over transformations is not a bound on real-task failure probability, and a proof about labels is not a proof of update sufficiency.

Statistical inference is likewise scoped. Identification bounds account for unresolved outcomes within the fixed sample. The small-stratum bootstrap is an approximate development-distribution diagnostic. Neither makes model backends, repeats, variants, or transformations into independent samples. The selected methods, incomplete selection history, missing deployed provenance, and unrun planned third model preclude a retrospective confirmatory significance claim.

10 Conclusion

The CCA v4 pilot shows why current correctness is not enough to evaluate compressed memory under updates. A matched-current pair can require different answers to the same future, and targeted deletion of older versions or tombstones destroys those distinctions in the mechanisms that need them. Both backends favor original frontier over the configured structured writer on a demanding joint endpoint, but that comparison includes delivery and answer-contract effects.

A fuller audit changes the scientific conclusion rather than merely adding caveats. It distinguishes 26 and 25 valid-but-wrong structured reveal logs from GLM’s 14 correct-value frontier formatting failures. It also exposes identifier-order dependence and provides a label-equivariant repair whose late-reference adequacy is lower on the original instances. Correcting a shortcut is not evidence of a stronger compressor. The defensible outcome is a reproducible, bounded methodology for evaluating update sufficiency and diagnosing its failure modes, with an explicit path to stronger prospective validation.

Ethics, data, and AI assistance

The analyzed tasks are synthetic event records, not observations of human participants. Distributed derivative receipts omit provider request identifiers and system fingerprints while retaining analysis-relevant content and provenance hashes. Applying persistent memory to people would require consent, access control, and deletion safeguards beyond this study; remembering invalidation does not authorize retaining personal data.

AI systems assisted research-design discussion, analysis-code development, artifact auditing, diagnostic analysis, the reference repair, and manuscript preparation. This assistance extended beyond language editing. No AI system is an author. Automated tests and multiple author-side revision passes are not independent peer review. The human author is responsible for checking the final manuscript, the provenance of the paid runs, release permissions, and any venue-specific disclosures. No unsupported funding or conflict-of-interest declaration is inferred from the supplied materials.

References

- Bhargava, Nishant, and Rodrigo Sobral Barrento. 2026. “MEMAUDIT: An Exact Package-Oracle Evaluation Protocol for Budgeted Long-Term LLM Memory Writing.” *arXiv Preprint arXiv:2605.02199*. <https://doi.org/10.48550/arXiv.2605.02199>.
- Chen, Tsong Yueh, Fei-Ching Kuo, Huai Liu, Pak-Lok Poon, Dave Towey, T. H. Tse, and Zhi Quan Zhou. 2018. “Metamorphic Testing: A Review of Challenges and Opportunities.” *ACM Computing Surveys* 51 (1): 4:1–27. <https://doi.org/10.1145/3143561>.
- Chhikara, Prateek, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. “Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory.” *arXiv Preprint arXiv:2504.19413*. <https://doi.org/10.48550/arXiv.2504.19413>.
- Efron, Bradley. 1979. “Bootstrap Methods: Another Look at the Jackknife.” *The Annals of Statistics* 7 (1): 1–26. <https://doi.org/10.1214/aos/1176344552>.
- Holm, Sture. 1979. “A Simple Sequentially Rejective Multiple Test Procedure.” *Scandinavian Journal of Statistics* 6 (2): 65–70. <https://www.jstor.org/stable/4615733>.
- Jin, Keyan, Yapeng Wang, Leonel Santos, Tao Fang, Xu Yang, Sio Kei Im, and Hugo Gonalo Oliveira. 2025. “Reasoning or Not? A Comprehensive Evaluation of Reasoning LLMs for Dialogue Summarization.” *arXiv Preprint arXiv:2507.02145*. <https://doi.org/10.48550/arXiv.2507.02145>.

- Kang, Minki, Wei-Ning Chen, Dongge Han, Huseyin A. Inan, Lukas Wutschitz, Yanzhi Chen, Robert Sim, and Saravan Rajmohan. 2025. “ACON: Optimizing Context Compression for Long-Horizon LLM Agents.” *arXiv Preprint arXiv:2510.00615*. <https://doi.org/10.48550/arXiv.2510.00615>.
- Lindenbauer, Tobias, Igor Slinko, Ludwig Felder, Egor Bogomolov, and Yaroslav Zharov. 2025. “The Complexity Trap: Simple Observation Masking Is as Efficient as LLM Summarization for Agent Context Management.” *arXiv Preprint arXiv:2508.21433*. <https://doi.org/10.48550/arXiv.2508.21433>.
- Littman, Michael L., Richard S. Sutton, and Satinder Singh. 2001. “Predictive Representations of State.” In *Advances in Neural Information Processing Systems*. Vol. 14. https://proceedings.neurips.cc/paper_files/paper/2001/file/1e4d36177d71bbb3558e43af9577d70e-Paper.pdf.
- Min, Guanghui, Liang Wu, Mayank Darbari, Chen Chen, and Liangjie Hong. 2026. “Toward Reliable Context Compression for Long-Horizon Agents: An Empirical Study of Execution Instability.” *arXiv Preprint arXiv:2608.06503*. <https://doi.org/10.48550/arXiv.2608.06503>.
- Montgomery, Jacob M., Brendan Nyhan, and Michelle Torres. 2018. “How Conditioning on Posttreatment Variables Can Ruin Your Experiment and What to Do about It.” *American Journal of Political Science* 62 (3): 760–75. <https://doi.org/10.1111/ajps.12357>.
- Rasmussen, Preston, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. “Zep: A Temporal Knowledge Graph Architecture for Agent Memory.” *arXiv Preprint arXiv:2501.13956*. <https://doi.org/10.48550/arXiv.2501.13956>.
- Trivedi, Harsh, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjana Balasubramanian. 2024. “AppWorld: A Controllable World of Apps and People for Benchmarking Interactive Coding Agents.” *arXiv Preprint arXiv:2407.18901*. <https://doi.org/10.48550/arXiv.2407.18901>.
- Wu, Di, Zixiang Ji, Asmi Kawatkar, Bryan Kwan, Jia-Chen Gu, Nanyun Peng, and Kai-Wei Chang. 2026. “LongMemEval-V2: Evaluating Long-Term Agent Memory Toward Experienced Colleagues.” *arXiv Preprint arXiv:2605.12493*. <https://doi.org/10.48550/arXiv.2605.12493>.
- Wu, Di, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2024. “LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory.” *arXiv Preprint arXiv:2410.10813*. <https://doi.org/10.48550/arXiv.2410.10813>.
- Xu, Wujiang, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. “A-MEM: Agentic Memory for LLM Agents.” *arXiv Preprint arXiv:2502.12110*. <https://doi.org/10.48550/arXiv.2502.12110>.
- Zerhoubi, Saber, Jelena Mitrovic, and Michael Granitzer. 2026. “The Compaction Cliff in Long-Running AI Agent Memory.” *arXiv Preprint arXiv:2608.22752*. <https://doi.org/10.48550/arXiv.2608.22752>.

Appendix A. Complete paid results and mechanism localization

Table A1. All original strict branch results. Every cell has 96 planned outcomes. Asterisk: one NA. C = current; R = reveal; O = override. Archive-access conditions must not be treated as storage-equivalent to strict bounded memory.

Condition	DS C	DS R	DS O	GLM C	GLM R	GLM O
Frontier	96/96	96/96	93/96	78/96	82/96	85/96
Latest-only	96/96	32/96	95/96	84/96	23/96	87/96
No tombstones	95/96	79/96	95/96	82/96	65/96	84/96
Structured H	76/96	62/96*	89/96	64/96	56/96	80/96*
Prose H	46/96	15/96	60/96	20/96*	11/96	30/96
Prose off	73/96	14/96	75/96	22/96	3/96	31/96*
Tail	24/96	4/96	0/96	16/96	4/96	0/96
Type-pin	24/96	16/96	0/96	19/96	10/96	0/96
Full archive	96/96	95/96	95/96	83/96	75/96	86/96
Closure archive	93/96	96/96	94/96	80/96	76/96	79/96
BM25 archive	79/96	35/96	96/96	60/96	29/96	83/96
Direct H archive	42/96	42/96	39/96	26/96	26/96	28/96

Table A2. All original joint scores. Every cell has 48 pair-repeat blocks within 24 source pairs. Asterisk: one unresolved block. Current outcomes are not part of this endpoint.

Condition	DeepSeek joint	GLM joint
Frontier	45/48	28/48
Latest-only	15/48	5/48
No tombstones	38/48	20/48
Structured H	19/48*	13/48*
Prose H	1/48	0/48
Prose off	1/48	0/48
Tail	0/48	0/48
Type-pin	0/48	0/48
Full archive	46/48	23/48
Closure archive	46/48	24/48
BM25 archive	17/48	8/48
Direct H archive	4/48	0/48

Table A3. Strict reveal counts by mechanism. F = frontier; L = latest-only; NT = no tombstones. Every cell has 16 planned evaluations but only four independent pairs. This is mechanism localization, not six natural-domain replications.

Mechanism	DS F	DS L	DS NT	GLM F	GLM L	GLM NT
Rollback	16/16	0/16	16/16	12/16	0/16	13/16
Expiry	16/16	0/16	16/16	15/16	0/16	14/16
Choice switch	16/16	16/16	15/16	13/16	11/16	12/16
Late reference	16/16	16/16	16/16	15/16	12/16	13/16

Mechanism	DS F	DS L	DS NT	GLM F	GLM L	GLM NT
Tombstone replay	16/16	0/16	0/16	15/16	0/16	0/16
Mixed	16/16	0/16	16/16	12/16	0/16	13/16

The latest-only condition’s twelve exact early-memory collisions do not account for all its failures: losing a dependency can invalidate an answer even when remaining irrelevant records distinguish the strings. Conversely, full history is an information-access reference, not a perfect LLM reader. Its occasional strict errors are consistent with adequate evidence and imperfect inference or output rendering.

Appendix B. Delivery, missingness, and cost accounting

Table B1. Distinct logical writer operations. The early write shared by current, reveal, and override is counted once. Raw-empty, over-cap, and incomplete columns can overlap. NA is not a zero-cost or semantic-failure label.

Backend	Writer	Done	NA	Empty	Over cap	Incomplete
DS	Prose H	480	0	115	0	121
DS	Prose off	480	0	0	8	0
DS	Structured H	478	1	28	0	35
DS	Direct H archive	288	0	157	0	162
GLM	Prose H	480	0	85	0	135
GLM	Prose off	480	0	0	7	0
GLM	Structured H	479	1	19	0	36
GLM	Direct H archive	288	0	47	1	68

For each source variant, repeat, and recursive LLM arm, the plan allows one early and four update writes. A direct archive writer allows one early and two final branch rewrites. Thus a recursive arm has at most 480 writer operations per backend and direct H at most 288. Across all conditions the ceiling is 1,728 writer plus 3,456 reader operations per backend, or 5,184. The count is not obtained by summing writer receipts copied into several branch rows.

DeepSeek’s missing structured reveal writer suppresses a later update and its reader. GLM has one missing structured override writer and two missing readers, on prose-off override and prose-H current. The missing GLM writer suppresses one downstream read. Exported descriptions of timeout or transmission failure do not replace absent raw error logs. The single metering-boundary response belongs to an already recorded DeepSeek prose-H override read, finishes by length, and has no visible answer. It remains a failed read under the unchanged contract.

The operation totals are consequently DeepSeek 5,181 done + one missing, and GLM 5,180 done + three missing. Recorded operations need not equal planned requests, and completed response counts need not equal all provider-billed attempts. Raw ledgers and invoices are required for the latter claim. No account prices or missing-call estimates are invented in the analysis.

Appendix C. Exact diagnostic taxonomy and repair contract

For each planned outcome, the classifier first preserves NA as unresolved. For an observed outcome it attempts to parse the memory as the public event schema and resolve the gold query. Unparseable

memory is a separate category. Valid but wrong event states are split according to strict reader success; valid correct states are split into strict pass, exact-wrapper-only failure, and other reader failure. This creates a disjoint partition, not independent percentages that can be added across overlapping writer diagnostics.

The parser and exact answer rules are intentionally conservative. It is possible for a nonconforming memory to support a correct reader answer. It is also possible for a malformed reader answer to reveal correct values. These cases are preserved rather than forced into a single semantic-loss label. Complete per-row assignments, pair contrasts, and leave-one-family-out bounds are distributed as CSV files.

Frontier-alpha uses the following strict-memory update. The capitalized steps refer to the public deterministic event semantics; they are not calls to a model or an external archive.

```
ALPHA(previous_memory, new_events, targets, budget):
  events = NORMALIZE(DECODE(previous_memory) + new_events)
  live, now = LIVE_VALUE_RECORDS_AND_MAX_CLOCK(events)
  controls = revocations + last clock attaining now
  closure = KEY_CLOSURE(targets, all live versions)
  stable-sort controls by is_clock, descending
  stable-sort live by (target / closure / other, revision), descending
  memory = empty event envelope
  for record in controls followed by live:
    if canonical UTF-8 envelope with record fits budget:
      append record
  return charged envelope; reject if even the empty envelope cannot fit
```

This tie-break uses the current retained order, not an uncharged original timestamp or hidden rank. Preserving it through the event array is necessary for the update contract. If equal-priority records are permuted, selection may change. If renaming changes escaped serialization size, capacity may change. Both cases are outside the proved label-equivariance premise.

Table C1. Offline late-reference budget sweep on the eight original variants. No model calls are involved; caps were varied retrospectively. These counts are neither independent test samples nor recommended optimal capacities.

Budget (bytes)	600	900	1,200	1,800	2,400	4,800
Original frontier	8/8	8/8	8/8	8/8	8/8	8/8
Frontier-alpha	2/8	2/8	2/8	2/8	4/8	8/8

The repair audit includes 23,328 state-evaluation rows (identity, renaming, permutation; two selectors; three branches) and 1,728 budget-sweep rows. The 13,440 stage-equivariance comparisons additionally check intermediate updates, reusing early states where branches fork. Model-call count and number of new independent tasks are both zero. No confidence interval is attached to repeated deterministic transformations.

Appendix D. Reproduction and artifact identities

The public-data SHA-256 is

ef7984bca91937f31ff6073d3431d0aed7421391e71376a860639cb6eed4a879.

The gold-data SHA-256 is

4bb8f2330378ee4a556691cf044fa50d6057b96bf12a31c30f5e895055c32236.

The uploaded paid-pilot archive SHA-256 is

d64e202c9e53dc5aa5d5e56bb9696e74f81a2ba7d805ce6db03db1e76e7cb610.

The data seed is 2026091101; run and bootstrap seed is 20260911. The matching dataset manifest is `cca4-matched-prefix-v1`, split `dev`, with 24 pairs, 48 variants, 32 distractors, and two update chunks. The distributed reference implementation is labeled as such; it is not a certification of the entire paid execution tree.

From the package root, `python analysis/run_all.py` regenerates the numerical analyses without model calls. `python -m unittest discover -s analysis -p 'test_*.py' -v` executes unit, metamorphic, accounting, and evidence-boundary tests. `python analysis/paper_assets.py` rebuilds the tables and figures from audited CSVs. Building PDF, DOCX, and LaTeX additionally requires Pandoc and the documented rendering dependencies. The release manifest verifies the distributed bytes before modification; it is an integrity check, not a trusted timestamp or third-party attestation.

The artifact includes a claim-to-evidence registry, the old-to-new revision record, the reference repair, a proposed confirmation contract, and a source-verification note. The contract is explicitly disabled for execution and cannot authorize paid experiments. Its purpose is to keep later empirical claims separate from this exploratory manuscript and to prevent retrospective development results from being relabeled as confirmation.